Presentation Video — narration, v4

iGEM Claremont-CA 2026 | draft of 2026-10-25

Timecodes assume 140 words per minute; the uploaded file has a hard limit of 15:00.

Line numbers run continuously — send edits back as *line 212: …*

Opening

120 words ends 0:51

ON SCREEN — Team logo, a team photo, then the title card.

We're iGEM Claremont-CA, a first-year team in California, and this is Phagenomics.

Pseudomonas causes serious infections in immunocompromised patients and persistent crop disease in agriculture, and phage therapy is one of the alternatives to antibiotics people are betting on. Its bottleneck is host range: each phage binds a specific receptor on a specific strain, and we can't design that specificity yet, because we don't understand how a phage recognises its target in the first place.

So we didn't set out to build a therapeutic. We set out to understand the recognition step any therapeutic would depend on. We went at it four times this season. Three of those attempts failed, and the failures turned out to be the useful part.

Act I.1 | The bottleneck

62 words ends 1:18

ON SCREEN — Plaque assay animation; then the same drop on two other strains, no effect.

A drop of phage goes onto a lawn of bacteria, and a few hours later there's

a clear circle where the bacteria used to be. Move the same drop to the next plate and nothing happens.

Same phage, different strain. That fussiness is the bottleneck: you need a phage that matches the strain in front of you, and usually there isn't one.

Act I.2 | Adsorption and the sugar coat 63 words ends 1:45

ON SCREEN — Cross-section of a cell: membrane, wall, O-antigen, capsule. Mark which layer the phage meets first.

It gets decided at the first step, before any DNA moves: the phage has to recognise the outside of the cell and stick to it.

And many bacteria aren't presenting a bare protein surface. They're wrapped in sugar — long chains called the O-antigen, and above that, in some species, a capsule. The first thing a phage touches is a forest of sugar.

Act I.3 | Where the project came from 61 words ends 2:11

ON SCREEN — Team photo with Prof. Libeskind-Hadas, captioned as an early conversation.

One thing belongs at the start, because it's why half this project exists.

Early on, Professor Ran Libeskind-Hadas at Claremont McKenna suggested reconstructing an ancestral phage — a whole organism, inferred from its relatives. We couldn't build one. But scale that down to a single enzyme and you can order it from a synthesis company. That is what this

project became.

Act I.4 | What we isolated

133 words ends 3:08

ON SCREEN — Sampling photo; five isolates collapsing to three strains; the circular phage genome with the tail spike outlined.

Soil from the California Botanic Garden, on the nineteenth of May. Out of it came five *Pseudomonas* isolates, whole-genome sequenced, all five closing into complete circular genomes. They collapse into three strains: two of them the same species, the third a different one.

That leaves two pairs of near-identical isolates, and those pairs do real work for the rest of the project. They're our internal control — any difference we report between strains has to be bigger than the difference we can measure within one.

And one phage, Phi-8, which infects exactly one of the three strains. Sixty-two thousand, three hundred and thirty-nine base pairs, circular, eighty genes, no match in the phage database, and effectively no genes in common with T7 or anything else characterised. The architecture is familiar. The sequence isn't.

Act I.5 | The first clue

173 words ends 4:22

ON SCREEN — The three cassettes as gene arrows, then the presence-absence matrix of the 45 clusters with the four shared columns marked.

Then we went to the locus that builds their sugar coat, anchored on the gene that flanks it in *Pseudomonas*, and compared all three.

The cassette sits in the same slot in every strain. The contents are unrelated: twenty-three genes in one strain, fifteen in another, twenty-five in the third, and of forty-five gene clusters across the three, only four are shared by all of them — all four generic sugar chemistry. No shared flippase, no shared polymerase, no shared chain-length regulator. Meanwhile the near-identical pairs inside each strain are a hundred percent identical across every cassette gene. Within a strain, zero variation. Between strains, almost total.

We also found that nobody could have told us this from a database: there is no serotype scheme for this group of species at all. The two standard tools are built for other organisms.

Three strains, three different sugar coats, one phage that kills one of them. So our first hypothesis: the sugar coat might be one of the main things setting what Phi-8 can attack.

Cycle 1 | Design

74 words ends 4:54

ON SCREEN — DBTL bar, DESIGN lit. Lock and key.

But that isn't the hypothesis we tested first.

The textbook picture of attachment is a receptor-binding protein on the tail meeting a protein on the bacterial surface. Lock and key. If that's what you assume, the obvious move is to work out which binding protein fits which receptor.

If we'd known the receptor was a sugar we'd never have run this. We

thought it was a protein, so we went looking for a protein.

Cycle 1 | Build

179 words ends 6:10

ON SCREEN — BUILD lit. Funnel from 441 to 144 clusters to 114 × 63. The surrogate-host trade-off as two columns.

First we needed to know what was on the surface. We annotated four hundred and forty-one candidate surface receptors across all five isolates and collapsed them into a hundred and forty-four clusters.

Then we needed a reference set of phages, which meant choosing a surrogate host — a relative of our strains that people have already characterised phages on. The closest match by receptor content had twelve phages on record. A slightly more distant relative had two hundred and twenty-seven binding-protein entries across seventy-one phages. We took the one with the literature rather than the one with the best identity, because a receptor shared with a strain nobody has studied tells you nothing.

That gave us a hundred and fourteen binding proteins against sixty-three receptors — seven thousand, one hundred and eighty-two pairs — each folded as a complex on our university's GPU cluster and ranked by the interface score.

Getting it to run took three tools: ColabFold throttled by its alignment server, Boltz-2 blocked by the cluster's CUDA version, and finally ESMFold, which was the one that ran.

Cycle 1 | Test

159 words ends 7:18

ON SCREEN — TEST lit. The ranking first so it looks reasonable, then score against protein length, then the control table. Hold 0.137 and 0.136 large.

Six thousand, three hundred and fifty-eight pairs came back with a score; the rest were too long for the model. And the ranking looked fine, which is the trouble with this kind of output — it always looks fine.

So before building anything on it, we ran nine control folds: three phage-receptor pairs known to bind, two of them with solved co-structures, each rerun with the receptor scrambled and mismatched against the wrong phage protein.

T5's pb-five with FhuA scored zero point one three seven. The same receptor scrambled scored zero point one three six. The other two pairs behaved the same way. None of the real binders separated from the controls.

And the scores tracked protein length, correlation of plus nought point six. The screen was ranking proteins by size.

We assumed that was the tool, so we rebuilt it with real evolutionary alignments. It got worse — under that one the mismatched pair outscored the genuine binders.

Cycle 1 | Learn

207 words ends 8:47

ON SCREEN — LEARN lit. Two confidence bars: on scrambling the shape score collapses 0.55 to 0.17, the interface score does not move.

Two tools, two control gates, both failed. One failure is about your run;

two independent failures are about the method.

The diagnostic is this. These models report two confidences: one for the shape of the molecule, one for how two molecules sit together. On the scrambled receptors the shape score collapsed, from zero point five five to zero point one seven — the model could plainly tell the sequence was scrambled. The interface score barely moved. It had never been reading an interface at all.

The reason is that these models don't simulate two proteins touching. They read evolution — looking across thousands of species for positions that change in step. If a residue on one side mutates and one on the other mutates to match, those two are probably in contact.

That signal only exists where two proteins have a shared history of co-adaptation. A phage protein and a bacterial protein have never co-adapted as a pair, so there's no history to read and the interface score returns noise.

So this was never a result about our proteins. It's a property of the method, and more compute doesn't fix it. Nine control folds decided the fate of seven thousand, and we never ran the real screen.

Cycle 2 | Design and Build

127 words ends 9:42

ON SCREEN — The classifier over 80 proteins; then the rung-period idea drawn on a coiled prism.

If we couldn't ask what the tail spike binds, we could ask what it is.

Two independent readouts, and they don't share a method. An open-

source sequence classifier over all eighty Phi-8 proteins. And then geometry: a right-handed beta-helix is a chain coiled into a long prism with the coils hydrogen-bonded in register, so if you measure how far apart in sequence each backbone hydrogen bond reaches and histogram it, a beta-helix peaks at its coil period, somewhere between sixteen and twenty-six residues. An alpha-helix peaks at four. A beta-sandwich has no peak at all.

We chose that test rather than fell back on it. On a genome with no close relatives, a database can't contain the answer — and homology annotation had already missed something here.

Cycle 2 | Test and Learn

221 words ends 11:16

ON SCREEN — The seven-protein verdict table, five of them turned down. Then the beta-helix render and a sugar-cutting animation.

Eight predicted structures covering seven proteins, each classified from its own backbone geometry. Five of the seven were turned down — and the periods they return aren't noise: an alpha-helical fibre lands at six, close to where an alpha-helix should be. The method says no, which is what makes it worth believing when it says yes.

It said yes twice. CDS 11, the tail spike, at classifier confidence nought point nine nine seven and a rung period of sixteen to seventeen: a right-handed beta-helix. And CDS 24, which the annotation had filed as a hypothetical protein, at nought point nine nine eight on a different period — a second depolymerase that homology-based annotation had missed

entirely on this novel genome.

That fold is a signature. Enzymes that chew through long sugar chains look like this, because a sugar chain is a thread and you need something that holds it straight while you cut. You can recognise a wood plane by its shape without being told what it cuts.

So the tail spike isn't a key. It's a blade — and the lock we'd spent months looking for doesn't exist.

That reading arrived from two directions at once: our classifier flagged it, and our PI reached the same conclusion from his own structural search. Between us we turned it into a hypothesis.

Cycle 2 | The hypothesis

44 words ends 11:35

ON SCREEN — One-sentence hypothesis box on white, held. Credited jointly with the PI.

In bacteria that wrap themselves in sugar, the ceiling on host range is set by what the depolymerase can digest — not by which protein the tail binds. And the narrow specificity we see now is a narrowed-down version of something broader further back.

Cycle 2 | The prediction, and its test

125 words ends 12:29

ON SCREEN — The 18-host matrix. Put all three caveats on screen with it: no no-phage control, one isolate unscored, two readouts never recorded.

That hypothesis makes a prediction you can falsify: if the tail spike reads Strain 2's sugar, Phi-8 should fail on the other two strains.

We ran it in July, eighteen host isolates against two pooled lysates. The Strain 1 and Strain 3 representatives scored negative against both. That is the direction the prediction makes.

We're not calling it confirmed, and the reasons are from our own notebook, not from a reviewer. The assay carried no no-phage control — there's a margin note on the page that just says *need control*. One isolate wasn't scored. And two follow-up readouts that would have settled it were never written down. The result is consistent with the prediction and nothing contradicts it, but the clean experiment hasn't been done.

Cycle 3 | Line one: the enzyme that reads the receptor

96 words ends 13:10

ON SCREEN — Domain map of CDS 11, and the Registry card for the submitted fragment.

If the depolymerase sets the ceiling, the test is whether that domain, on its own, can cut the sugar.

CDS 11 is nine hundred and forty-one residues and most of it isn't the enzyme. The first hundred and ten anchor it to the virion. The next two hundred are a shaft that moves between every model we ran. The rigid trimeric body — from residue three hundred to the end, agreeing between models to within one ångström — is the part that carries the chemistry. That's the fragment we designed, expressed, and submitted to the Registry.

Cycle 3 | Line two: why the ancestors are a different protein

230 words ends 14:48

ON SCREEN — The judge will ask this. CDS 11 and CDS 12 side by side, captioned a separate protein, not a domain.

The ancestors are a different protein, and it's worth being clear about that.

Phi-8 also carries CDS 12, a glycoside hydrolase from family five. It's a TIM barrel, not a beta-helix, and its closest structural match is an endoglucanase caught with a short cellulose chain in its active site. It is a separate protein — not a domain of the tail spike — and what it does inside the phage is not something we have established.

So why reconstruct ancestors of that one, and not of the tail spike we've just spent five minutes on? Three reasons, and none of them is that we got confused. CDS 12 has a thousand homologues in the sequence databases, which is what you need to build a tree and reconstruct along it; the tail spike is novel enough that there isn't one. CDS 12 is a third of the size, so it was by far the better bet to express. And we were working inside a sponsored synthesis allowance, where four short constructs were affordable and four long ones were not.

It means the ancestral panel tests the shape of the hypothesis — does an ancestral glycoside hydrolase have a broader substrate range than its descendant — on an enzyme we can actually build, rather than on the one that reads the receptor. That's a real limitation and it's ours, not a reviewer's.

Cycle 3 | Build

159 words ends 15:57

ON SCREEN — Tree collapsing to three nodes; Registry cards for the six parts, published and CC0.

We worked out the direction with our PI; he ran the alignment, the tree and the reconstruction on sequences we prepared. Every node on the path from CDS 12 back to the root was reconstructed, and the three we picked were ordered as DNA.

Six constructs in all, in pET-28a with an N-terminal His-tag, synthesised by Twist across two orders — and all six are now published basic parts on the iGEM Registry under a CC0 licence. Three of the six are those reconstructed ancestors: protein sequences that exist nowhere in nature. That route to new parts is reusable by any team working on an enzyme family, so we wrote the method out in full.

One warning we'd pass on, because it cost us. Our first order was codon-optimised and our second wasn't — the second set carries a mammalian-like codon bias, which in an *E. coli* host is the first thing to suspect when a band doesn't appear.

Cycle 3 | What the model said, and its own control

145 words ends 16:59

ON SCREEN — ZymCTRL perplexity along the path, with the 21-of-24 control laid over it. Caption: perplexity is not activity.

Before any of them reached a bench, we asked a protein language model what the reconstructions looked like. Scored against twenty-four glycoside-hydrolase classes, the one it likes best is endo-one-four-beta-glucanase

— the same chemistry the structure points at. And along the path to the root the score improves steadily, which read on its own is the hypothesis coming true.

Except that it doesn't survive its own control. A reconstructed ancestor is consensus-like by construction, so it scores better under almost any label — and it does: the score improves in twenty-one of the twenty-four classes. Direction alone proves nothing. What survives is that the improvement is largest under the two labels the structure points at, and even that is less clean than one number would suggest.

And perplexity isn't activity. Nothing in that analysis measures substrate range, which is the whole point of the panel.

Cycle 3 | Test

141 words ends 17:59

ON SCREEN — Four test rounds: the 9-15 gels, the 9-29 band, the 10-01 elution that worked and the one 29 kDa short, the 10-10 column that did not.

Four rounds at the bench, and they didn't line up neatly.

September the fifteenth: two gels, four constructs, induced against uninduced. No detectable induction on either. The tail spike lane was essentially blank, which points at lysis or loading rather than at expression.

September the twenty-ninth: CDS 12 gave the first induction-dependent band at the right size.

October the first: nickel columns on six constructs, and one of them worked — the first ancestor eluted at its predicted mass, the only construct that did. The tail spike eluted something, but twenty-nine kilodal-

tons short of what it should be.

October the tenth: the same column on a second ancestor, and both elutions came off blank while the load and the flow-through carried the same band. We read that as a purification we'd botched, and were about to spend another week on it.

Cycle 3 | Learn: an outside view changed a result 240 words ends 19:42

ON SCREEN — The strongest human-practices beat. Tolstorukov's email, then the 10-19 dot blot with 2.9× marked.

What stopped us was a conversation.

On October the thirteenth we took it to Professor Ilya Tolstorukov at Keck Graduate Institute, who works on protein expression. He wrote back before we met to say he wasn't sure he was the right person — expression is his field, purification isn't — and met us anyway. That turned out to be the point.

He changed the question. Not *why is nothing eluting*, but *is there anything there to elute*. Check whether the protein is being made at all — a dot blot against an anti-His antibody — before touching the column again. He also told us to run uninduced controls, and to standardise how much material goes onto a membrane before comparing spots on it.

Six days later we ran that blot. The flow-through from the second ancestor read two point nine times the negative control.

So the protein was there. It had been made, it came off in the flow-

through, and the run failed at capture — not at expression. A gel can't tell those apart: no expression, insoluble protein, and a column that didn't bind all look identical. One cheap assay separates them, and it pointed the fix at the column instead of the construct.

We'd spent a week debugging the last step of a workflow without checking the first step worked, because the last step was the one visibly failing. That's the generalisable mistake, and one meeting fixed it.

Cycle 3 | The activity assay, and what it carries 169 words ends 20:54

ON SCREEN — The 10-16 spot test and colony counts against the PBS control. All three caveats on screen.

And then the assay the whole project points at: does an enzyme, on its own, do anything to the sugar?

We read it on October the sixteenth. No visible lysis on any plate — including Phi-8 on its own hosts, so the assay wasn't reading clearing at all. But counting colonies inside the drops gives a number. Under the first ancestor, across seven strains, colony counts fall to a median of twelve percent of the plate's PBS control. Phi-8's own median is zero.

Our PI reads that suppression as enzyme activity. We'd want more before we call it that ourselves, and there are two specific gaps. The control that separates protein activity from phage carried over into the preparation was not run. And the dilutions used in that spot test later read at or below the detection floor of our own dot blot — so the protein in those

drops was never actually shown to be there.

It's the strongest activity signal we have, and it is one plate.

Cycle 3 | What we still owe

48 words ends 21:15

ON SCREEN — Honest list.

And the gap that runs through all of it: eight protein assays now — seven gels and a dot blot — and not one of them carries a lane of untransformed BL21. Every round was designed around a different variable instead. That control is on the bench now.

Cycle 4 | Design and Test

191 words ends 22:37

ON SCREEN — Picks from the plaque zone, sequencing, 70 candidate sites down to one, F126L in a 30 kb capsule cluster.

Three cycles in, every guess about the receptor had come from the phage's side. So for the fourth we let the bacteria answer.

In July we picked colonies from inside the plaque zone — cells still growing where the phage should have killed them — and in October sequenced one from each host against its parent. No structure prediction anywhere in this loop.

Both picks turned out to be the parent strain, intact, with no phage sequence in them, so neither had simply become a lysogen. A naive assembly comparison gives sixteen to twenty-one differing sites, which means nothing on its own — two sequencing runs of the same clone already

differ by more than that. So every candidate went back to the raw reads: seventy sites examined, against the parent's reads and an independent read set with completely different chemistry.

Exactly one survived. A missense change in a Stealth-family sugar phosphotransferase, carried by two hundred and sixty-three of two hundred and sixty-four mutant reads and none of the parent's. It sits inside a thirty-kilobase cluster that builds and exports capsular polysaccharide. The other pick has no confirmed mutation at all.

Cycle 4 | Learn

112 words ends 23:25

ON SCREEN — A capsule hiding an intact O-antigen: how both conclusions hold at once.

That disagrees with Cycle 2, which is the whole reason for doing it this way round. Cycle 2 said the receptor is a sugar. Cycle 4 agrees it's a sugar and disagrees about which one — it points at the capsule, with the O-antigen cassette intact.

Both can be true at once, because a capsule sits on top and can hide an intact O-antigen underneath it. We don't know which surface Phi-8 reads, so where the distinction matters this project says *surface polysaccharide* and leaves it there.

And the honest gap: those colonies were never confirmed resistant. They were colonies in a plaque zone, which is suggestive and isn't the same thing.

What we're leaving behind

210 words ends 24:55

ON SCREEN — Tool output, the GitLab repository, the six parts, the negative result.

Four things, and one of them is a failure.

Two packaged tools, MIT licensed, on iGEM's GitLab. One does ladder calibration and densitometry on a gel photograph — we re-typed that code three times across three gel days before admitting we should generalise it. The other is the fold test from Cycle 2, which takes a single predicted structure and returns a call with no database behind it; it also tells a terminal repeat from a circular join by the shape of the read coverage, and that changed our own answer. This wiki used to say the genome was linear. The check said circular, and circular is what we report. Two more tools are staged with porting notes.

Six parts on the Registry, three of them sequences that never existed.

A benchmark: on a genome with no close relatives, a machine-learning classifier found a depolymerase that homology-based annotation had filed as hypothetical. Our recommendation for novel phage genomes is to run both and reconcile the disagreements.

And the screen that failed — published with the controls that caught it and the reason it could never have worked. Don't use interface scores to screen phage-host protein pairs, and run a positive control with a solved co-structure before you scale to thousands.

Coda | The garden

209 words ends 26:24

ON SCREEN — Interview clips, at least 20 seconds. Credit both hosts and note permission.

Two days before our wiki froze we went back to the garden the soil came from, and asked whether a treatment as strain-specific as ours would be any use to them.

Most of what we'd assumed was wrong. We expected to argue for biological control against a chemical default, and didn't have to — they already suppress soil fungi by mulching. We expected the objection to be ecological risk. It wasn't; it was whether the thing works. Copper is the only chemistry they have for bacterial disease, it goes onto every blossom every four or five days, and they have eighty-six acres and eight staff. The objection is arithmetic.

And the finding that costs us most: they've never had a sample come back positive for *Pseudomonas*. We isolated a soil phage and its environmental host — not a treatment for a disease this stakeholder has.

What they do need is something for bacteria inside the vascular system, where nothing they have reaches. Our strain doesn't solve that. But the question we worked on — what a tail spike reads on a cell surface, and how far that can be predicted from a genome — is the one you'd have to answer first. The strain is local. The question isn't.

Coda | Safety

126 words ends 27:18

ON SCREEN — The IBC letter, 0 VFDB / 0 CARD, and the open condition quoted.

On safety: our institutional biosafety committee gave conditional approval with one condition still open, and that condition is a question about host range — the subject of this entire project. They asked us to show our reasoning that the work won't alter host range, virulence, or antibiotic resistance. We don't have full approval; the revision is being written.

Phi-8 carries no virulence factors and no resistance genes across all eighty coding sequences, and our constructs are protein fragments in lab *E. coli* — nothing here is a modified phage. But understanding what sets the ceiling on host range is the same as knowing what would have to change to raise it, so we treat this as dual-use rather than arguing our way to a clean no.

Coda | Where we are

214 words ends 28:50

ON SCREEN — Three columns fading in: Established / Partly tested / Open.

So, where we actually are.

Established: five bacterial genomes and a novel circular phage genome; three unrelated sugar cassettes in the same locus, with no serotype scheme in existence to have told us; a tail spike that is a sugar-cutting enzyme by two independent methods; a controlled negative result about interface prediction; and one resistance mutation verified in the reads.

Partly tested: that Phi-8 won't infect the other two strains — the July assay pointed the right way and carried no control. And the enzyme

panel: one ancestor purified at its predicted mass, colony counts under it fall to twelve percent of control, and we won't call that activity without the control that separates it from carried-over phage.

Open: a clean expression round with the lane we keep not running; whether the tail spike fragment is even full length; which sugar layer the phage reads; whether Phi-8 is temperate; and depositing the genome, which we haven't done.

Four cycles, three of them wrong, each wrong in a way that told us where to go next. If there's one thing we'd pass on: put the cheap control in front of the expensive experiment. Nine folds told us what seven thousand couldn't, and one afternoon's dot blot told us what a week on a column hadn't.

4038 spoken words. With the act cards and a 30-second credit card, the finished file runs about 29:50 at 140 words per minute.